



API SECURITY RISK ANALYSIS



Presented for :
FUTURE INTERN

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1.Objective

The objective of this task is to perform a read-only security risk analysis of a public API, identify common API security weaknesses, and provide clear remediation recommendations in simple business language.

2.APITested

Base URL: <https://jsonplaceholder.typicode.com>

Endpoints Analyzed:

- /users**
- /posts**
- /posts/{id}**

3. Tools Used

- Postman (API testing and request inspection)**
- Browser DevTools (Header and response inspection)**
- Manual security analysis**

4. Findings and Risk Analysis

4.1 Open / Unauthenticated Endpoints

Observation: API endpoints were accessible without authentication.

Risk Level: Medium

Business Impact: Unauthorized users may access exposed data.

Remediation: Implement authentication using API keys, JWT, or OAuth.

4.2 Excessive Data Exposure

Observation: The /users endpoint exposed user information without restriction.

Risk Level: Medium

Business Impact: Sensitive data leakage may occur.

Remediation: Limit returned data and apply least-privilege design

4.3 Broken Object Level Authorization (BOLA)

Observation: Individual objects were accessible by modifying ID in the URL

(/posts/{id}).

Risk Level: Medium

Business Impact: Users may access other users' data.

Remediation: Enforce server-side authorization checks for each request

4.4 Missing Rate Limiting

Observation: No rate limiting or request throttling was observed.

Risk Level: Low to Medium

Business Impact: API abuse or automated scraping may occur.

Remediation: Implement rate limiting and request throttling mechanisms.

4.5 Missing Authentication / Authorization Headers

Observation: No authorization mechanism detected in API responses.

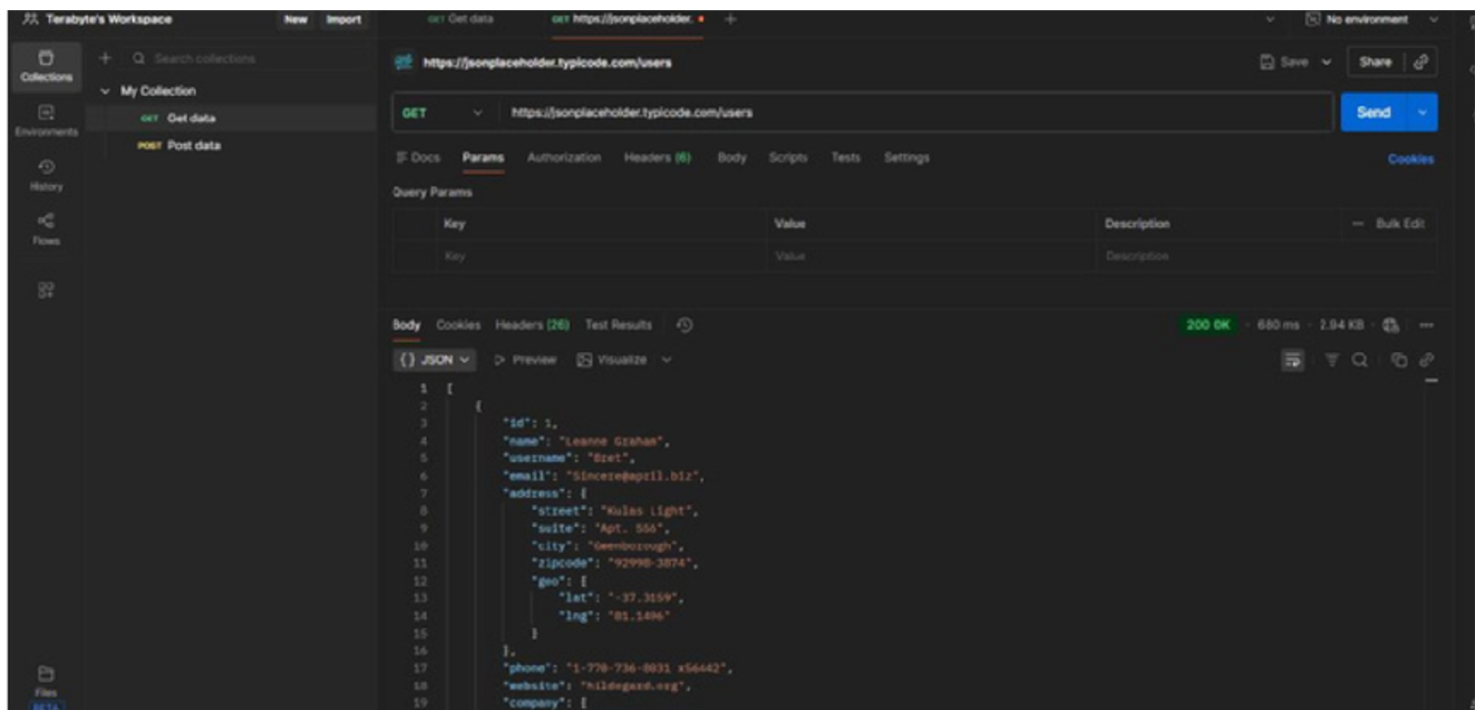
Risk Level: Medium

Business Impact: Unrestricted API access.

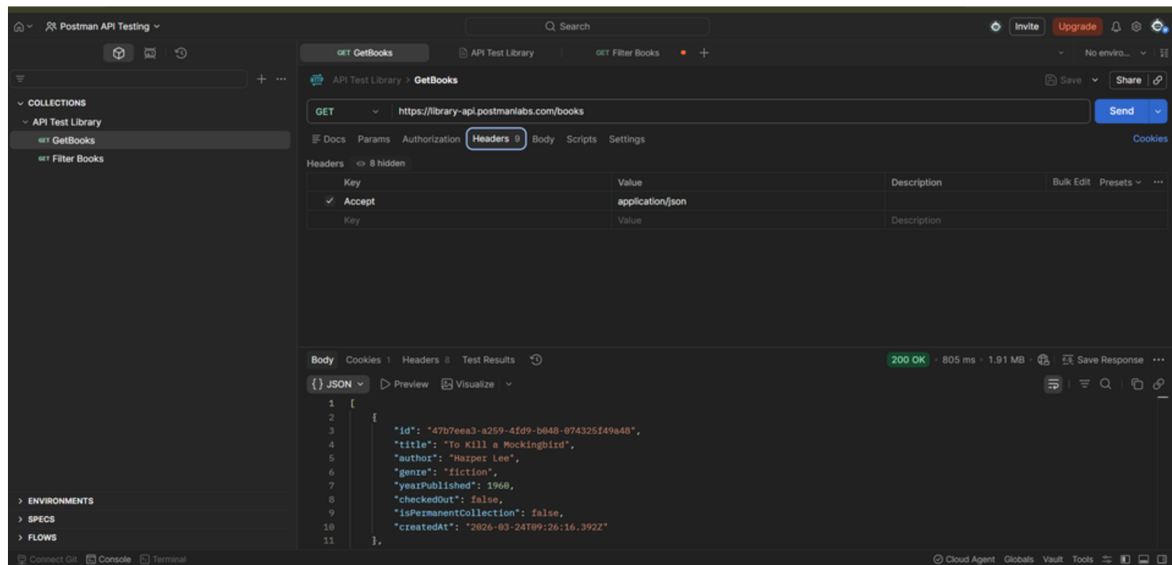
Remediation: Use secure token-based authentication and authorization controls.

5. Screenshots / Evidence

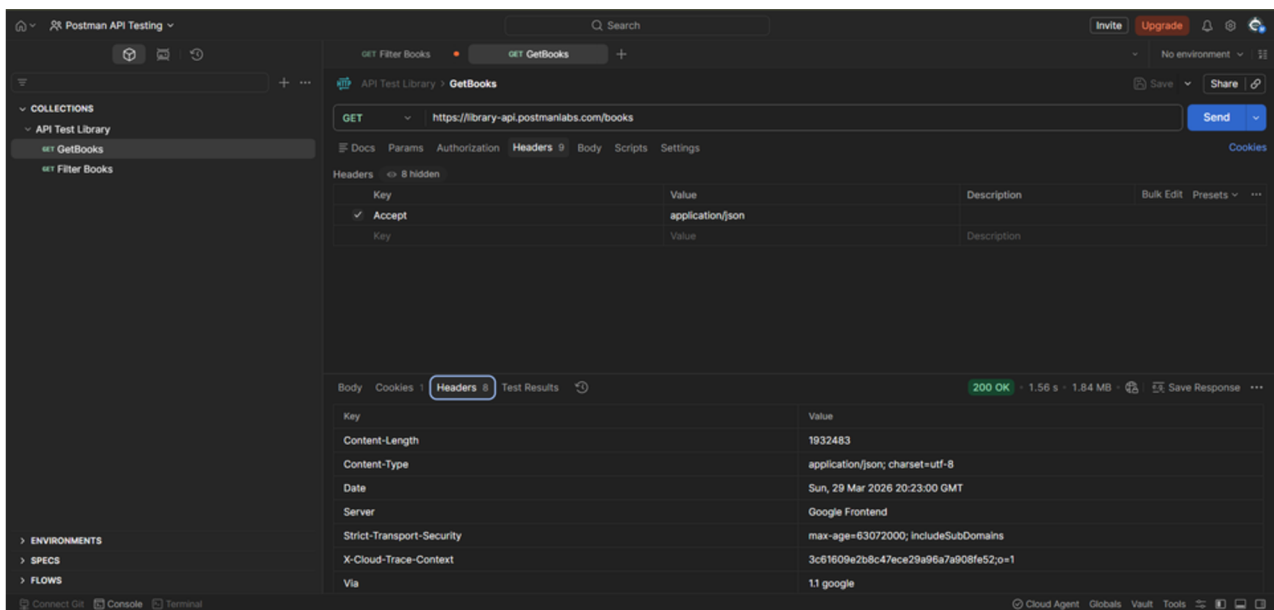
1.Usersendpointrequestand response



2. Posts endpoint request and response:



3. Header analysis screenshot:



6. Overall Risk Summary
High Risk: 0 Medium Risk: 4
Low Risk: 1

7. Conclusion

The API demonstrated multiple common security weaknesses including open endpoints, lack of authentication, and missing rate limiting.

Implementing proper authentication, authorization, and request controls would significantly improve the API security posture.